
State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

SURVEY

23,262 developers worldwide

SOURCE

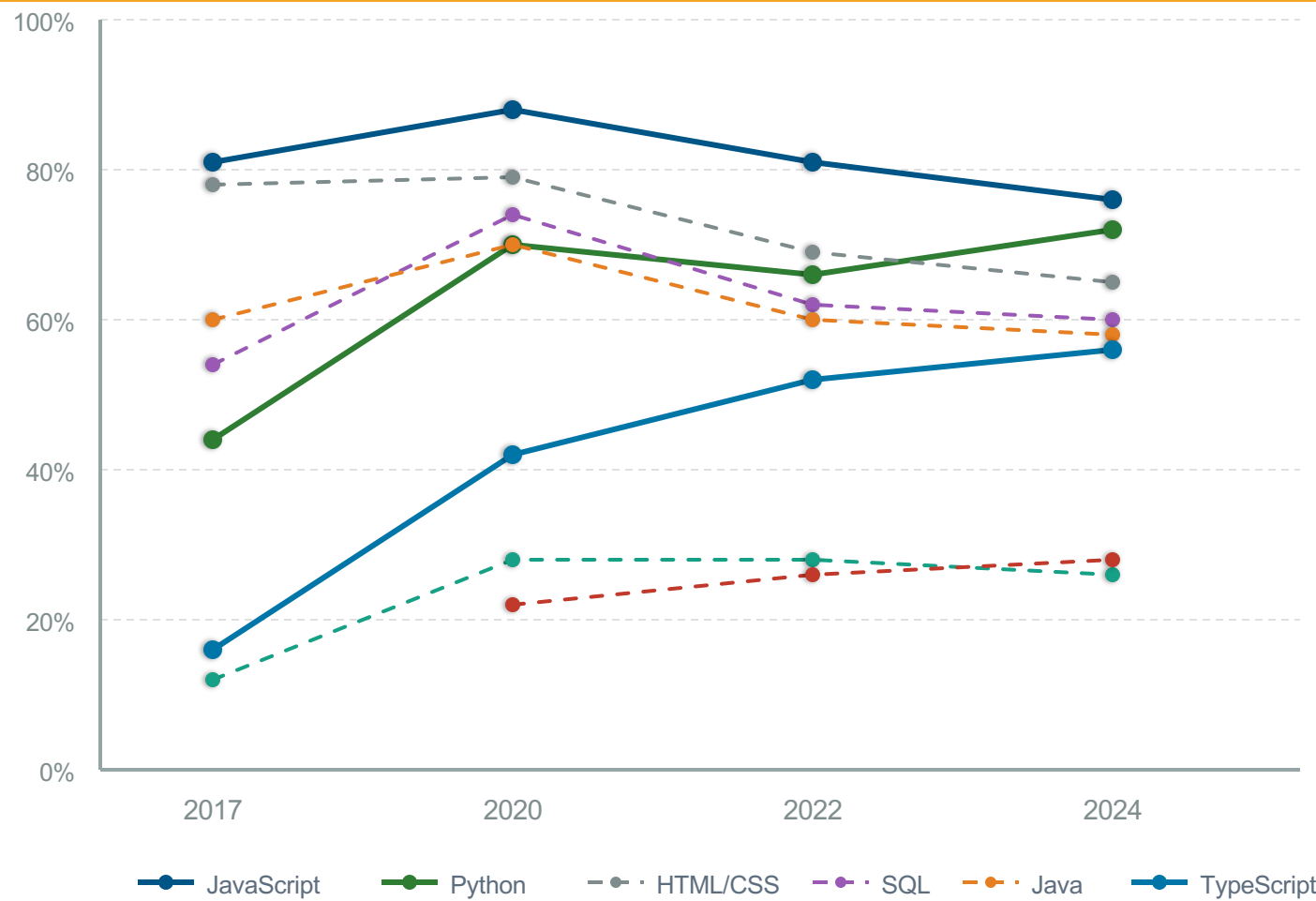
JetBrains DevEcosystem 2024



CONFIDENTIAL

KEY INSIGHT

JavaScript peaked at 70% in 2020 but declined to 61%; Python rose from 32% to 57%; TypeScript tripled from 12% to 37%



Usage Data (2017-2024)

Language	2017	2020	2022	2024
JavaScript	65%	70%	65%	61%
Python	32%	55%	53%	57%
HTML/CSS	60%	61%	54%	51%
SQL	42%	56%	49%	48%
Java	47%	54%	48%	46%
TypeScript	12%	28%	34%	37%
Go	8%	19%	19%	18%
Rust	—	7%	9%	11%

Growth leaders:

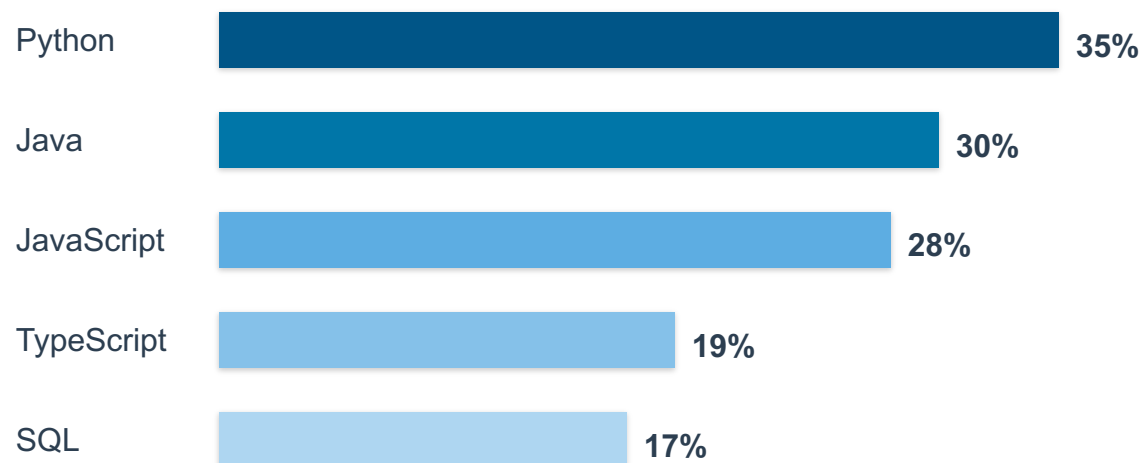
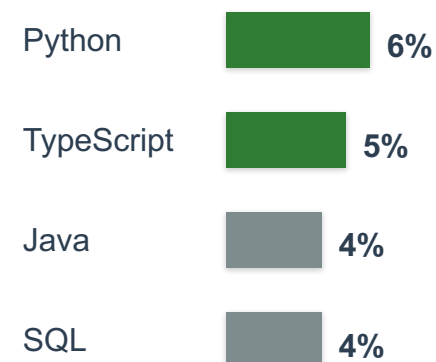
Python +25pp, TypeScript +25pp, Rust +4pp

Declining:

JavaScript -4pp, HTML/CSS -9pp, Java -1pp

KEY INSIGHT

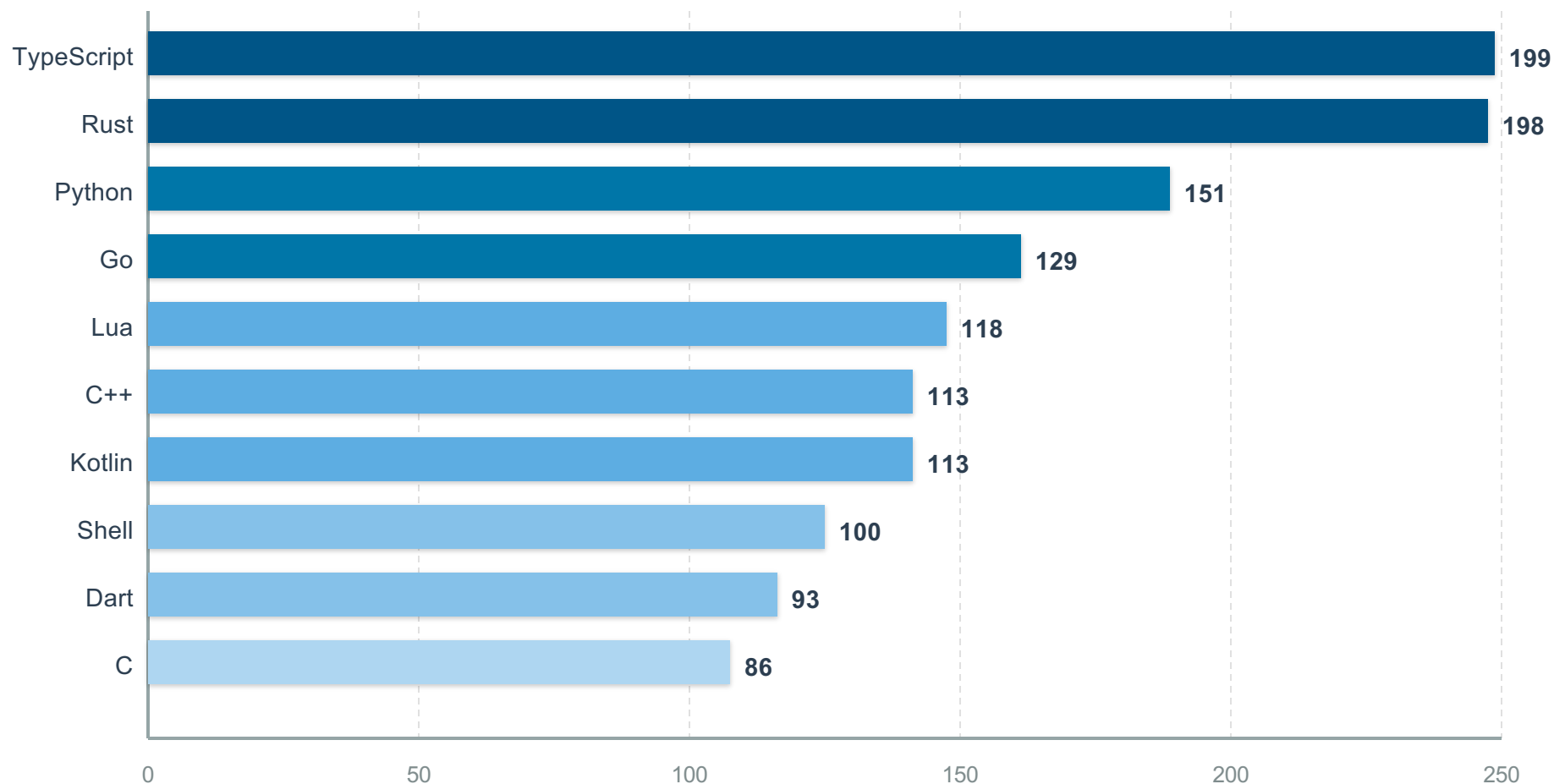
Adoption intent for new primary languages remains below 6% — developers stick with what they know

Primary Language Share**Adoption Intent (Plan to Adopt)****The Comfort Zone Effect**

Developers show strong loyalty to their chosen ecosystems. Despite Python's dominance as a primary language, only 6% plan to adopt it — suggesting ecosystem lock-in and expertise investment keep developers loyal.

KEY INSIGHT

Type-safe, modern-tooling languages dominate the top spots — satisfaction correlates with developer retention



KEY INSIGHT

Language choice is driven by application domain, not general preference — hire for the domain, not just the language

AI / Machine Learning

Python

31% of developers

Data Science / Analytics

Python

28% of developers

User Interface

TypeScript

62% | JS 58%

APIs & Services

Java

49% of developers

Application Logic & Workflows

Java

55% of developers

System Tools & Components

C++

27% of developers

Data Processing & Analytics

Python

17%

Integrating with APIs & Services

Python

38%

DOMAIN-DRIVEN HIRING

Python dominates AI/data (3 domains).

Java owns backend services. TS/JS controls the frontend.

KEY INSIGHT

Data-driven investment priorities: back the rising stars, modernize legacy stacks, and hire domain-first

01

Invest in TypeScript

Usage tripled from 12% to 37% in 7 years. Highest satisfaction (199).

→ **Adopt as primary frontend stack**

02

Double Down on Python

Dominates AI/ML (31%), data science (28%), APIs (38%). #1 primary.

→ **Core for AI/data initiatives**

03

Bet on Rust

Near-perfect satisfaction (198), strong in systems/tools. Rising fast.

→ **Pilot for performance-critical systems**

04

Modernize Legacy Stacks

PHP dropped 30%→17%, C# flat at 22%. Low satisfaction for legacy languages. Plan gradual migration to modern alternatives.

→ **PHP→TypeScript/Go, C#→Java/Go where possible**

05

Hire Domain-First

Language choice is domain-driven. Developers rarely switch primary stacks (<6% intent). Hire for the problem domain, not just language skill.

→ **AI team → Python experts; Frontend → TS/JS**